

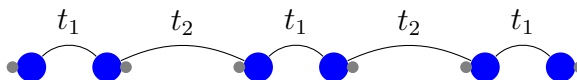
Homework 8

The Last One

1. An electron is moving in a one-dimensional chain. The chain is represented by a weak periodic potential associated with a lattice spacing a . An electric field E is applied along the direction of the chain. Assume a very long relaxation-time τ and very low temperature. Calculate the fundamental frequency of the Bloch oscillation for $a = 50\text{\AA}$ and $E = 5 \times 10^4 \text{ V/cm}$. Compare the result to the typical relaxation-time of electrons in a metal.
2. This problem explores the Peierls Instability. Assume a one dimensional chain of atoms with spacing a and having one electron per site. The “elastic” energy can be described by a nearest neighbor spring γ . Assume that the valence electrons are described by a nearest neighbor hopping parameter which vary linearly with distance $t = t_o + g(a - |x_{i+1} - x_i|)$. The valence band structure will change as a function of atomic displacement, so you will need to compute it and add it to the elastic energy in order to get the total energy.

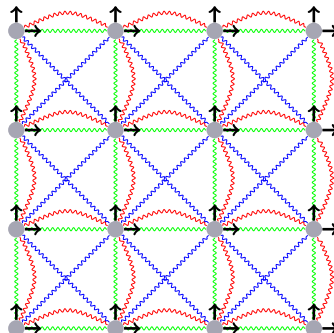
$$E = \sum_i \frac{1}{2} \gamma (x_{i+1} - x_i)^2 + \int \epsilon_k \theta(\epsilon_k - \epsilon_{Fermi}) dk$$

where ϵ_k is a function of the nuclear positions $\{x_i\}$. Evaluate the total energy as a function of the $q = \frac{\pi}{a}$ vibrational mode. The vibrational mode is shown below, with the grey points being the undistorted lattice.



Assuming that each atom displaces by a distance of $|\delta|$ from a lattice point, then we have $t_1 = t_o + 2g|\delta|$ and $t_2 = t_o - 2g|\delta|$. Choose values for the independent parameters and plot the total energy along the distortion.

3. Consider the square lattice linked by springs; the energy as a function displacement is given by the potential V shown below.
 - (a) Construct the dynamical matrix.
 - (b) Assuming $\gamma_\sigma = 1, \gamma_\pi = \frac{1}{3}, \gamma_d = \frac{1}{10}$, plot the vibrational frequencies for the k-space path $(0,0) \rightarrow (\frac{1}{2},0) \rightarrow (\frac{1}{2},\frac{1}{2}) \rightarrow (0,0)$.



$$\begin{aligned}
V = \sum_{ij} & \left[\frac{1}{2} \gamma_{\sigma} \left(u_{(i+1,j)}^{(x)} - u_{(i,j)}^{(x)} \right)^2 + \frac{1}{2} \gamma_{\pi} \left(u_{(i+1,j)}^{(y)} - u_{(i,j)}^{(y)} \right)^2 \right. \\
& + \frac{1}{2} \gamma_{\pi} \left(u_{(i,j+1)}^{(x)} - u_{(i,j)}^{(x)} \right)^2 + \frac{1}{2} \gamma_{\sigma} \left(u_{(i,j+1)}^{(y)} - u_{(i,j)}^{(y)} \right)^2 \\
& \left. + \frac{1}{2} \gamma_d \left(u_{(i+1,j+1)}^{(x)} + u_{(i+1,j+1)}^{(y)} - u_{(i,j)}^{(x)} - u_{(i,j)}^{(y)} \right)^2 + \frac{1}{2} \gamma_d \left(u_{(i+1,j-1)}^{(x)} - u_{(i+1,j-1)}^{(y)} - u_{(i,j)}^{(x)} + u_{(i,j)}^{(y)} \right)^2 \right]
\end{aligned}$$